Natalie Truesdell

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SUMMARY

Data Scientist with hands-on experience leading multimodal machine learning projects — from deep learning pipelines integrating imaging and clinical data to LLM-powered retrieval systems and interactive dashboards. Skilled in Python, PyTorch, and scalable cloud ML workflows, with a strong foundation in statistics and model evaluation. Experienced communicating technical work to both technical and non-technical stakeholders.

SKILLS

Languages, Databases & Tools: Python, SQL, R, Bash, Git
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, PyTorch, predictive modeling, feature engineering, supervised and unsupervised learning, deep learning, model evaluation, cross-validation, hyperparameter tuning
LLM Technologies: OpenAI API, LangChain, FAISS, Sentence Transformers, RAG pipelines, prompt engineering, vector search, embedding models, tool calling, LLM evaluation, external API integration
Data Engineering & Cloud: GCP (Vertex AI, GCS, BigQuery), AWS, DuckDB, PyArrow, CI/CD, GitHub Actions, data pipelines, workflow automation, data validation, QA and testing
Statistics: Hypothesis testing, regression analysis, Bayesian inference, experimental design, causal inference, A/B testing, time series analysis
Data Visualization & Reporting: Matplotlib, Seaborn, Tableau, Shiny, PowerPoint, Excel

EDUCATION

University of British Columbia, Vancouver (UBC) | *Master of Data Science* Aug 2025 – Jun 2026
Average: 92% (GPA: 4.33/4.33)

University of California, San Diego (UCSD) | *B.S. Business Economics, Minor in Data Analytics* Sept 2020 – Jun 2025
GPA: 3.6/4.0

EXPERIENCE

Data Analytics Intern | *Acorn Evaluation, Inc.* Jan 2025 – Jul 2025

- Developed and deployed a web scraping and parsing framework to identify potential clients across **200+ websites**, designing **custom extraction logic (XPath, regex)** and reducing manual research time by **~90% per project cycle**
- Built Tableau dashboards for **80% of the firm's client portfolio**, translating fragmented program data into actionable performance insights that improved client reporting efficiency and supported data-driven decision-making
- Leveraged and refined **ETL workflows** using Python and Power Query to extract, transform, and validate multi-source program data (30K+ records), improving **data consistency** and reducing manual errors
- Conducted data reconciliation across raw datasets to build a **validated master directory**, which served as the foundation for an automated web scraping pipeline that scaled lead discovery

Accounting Intern | *Lohman & Associates* Jun 2024 – Nov 2024

- Managed data operations for 90% of the firm's client portfolio across multiple platforms (QuickBooks, Harvest, Excel) to ensure billing accuracy and maintain reliable financial records
- Created weekly operational reports consolidating data from **20 client projects**, providing leadership with visibility into revenue forecasts, project performance, and resource allocation

PROJECTS

Multimodal AI for Clinical Risk Prediction | *Project Lead, Tandem Research & UBC Capstone*

- Developed an end-to-end predictive modeling pipeline integrating 3D CT imaging with EHR data from 23K patients to predict mortality and hospital readmission using **ResNet V2, LightGBM,** and **a Transformer-based model**
- Built a scalable cloud-based ML workflow on **GCP** using **DuckDB, BigQuery**, and **Vertex AI** to accelerate experimentation, improve reproducibility, and support patient-level prediction analysis
- Designed a **cascaded multimodal fusion approach** that improved 6- and 12-month risk prediction performance by an average of 5%, demonstrating the value of combining imaging and clinical data sources
- Implemented a comprehensive **explainability framework** utilizing TreeSHAP, Grad-CAM, and censoring experiments to quantify clinical feature importance and highlight the image regions driving predictions

FashionRAG: LLM-Powered Product Discovery Engine

- Designed a complete RAG system using 2.5M+ Amazon reviews, **integrating BM25 lexical retrieval, semantic embeddings, LangChain orchestration, and Llama-3.1-8B-Instant** with grounding constraints and source attribution to generate reliable product recommendations
- Improved recommendation quality through field-weighted representations, hybrid search fusion, and MMR reranking, increasing retrieval diversity while preserving relevance

- Developed a **qualitative and quantitative evaluation framework** using retrieval metrics and manual relevance analysis to benchmark search strategies and guide the selection of a hybrid architecture for improved query coverage
- Engineered a scalable data pipeline with **DuckDB**, vectorized preprocessing, and stratified sampling, reducing popularity bias in training data and achieving **3–5× faster corpus preprocessing times**

Geospatial Behavioral Analytics Dashboard — Dashboard

- Developed an interactive geospatial dashboard with an LLM-powered natural language interface (**QueryChat + GPT-4.1**), enabling non-technical users to explore geospatial data through conversational queries
- Designed a responsive visualization architecture using **Shiny for Python, Altair, and Folium**, ensuring synchronized state management across dynamic maps, custom HTML legends, and multi-metric charts
- Engineered a high-performance backend using **DuckDB** and **Parquet** to query over 3,000 Central Park squirrel observations in under a second, reducing query latency by 10%
- Implemented a **robust testing suite using pytest and Playwright** to automate UI verification, ensuring application reliability and consistency across complex user interactions

SurveyCleaner: Production Data Validation and Cleaning Package — TestPyPI

- Engineered a Python data validation package with a reusable API to standardize survey preprocessing workflows including deduplication, binary normalization, and ordinal encoding
- Adopted a **test-driven development (TDD) approach** to achieve 100% test coverage with pytest, ensuring reliable validation logic across all edge cases
- Established **fully automated CI/CD pipelines** using GitHub Actions, streamlining the full development lifecycle from automated testing and versioning to TestPyPI distribution and documentation hosting